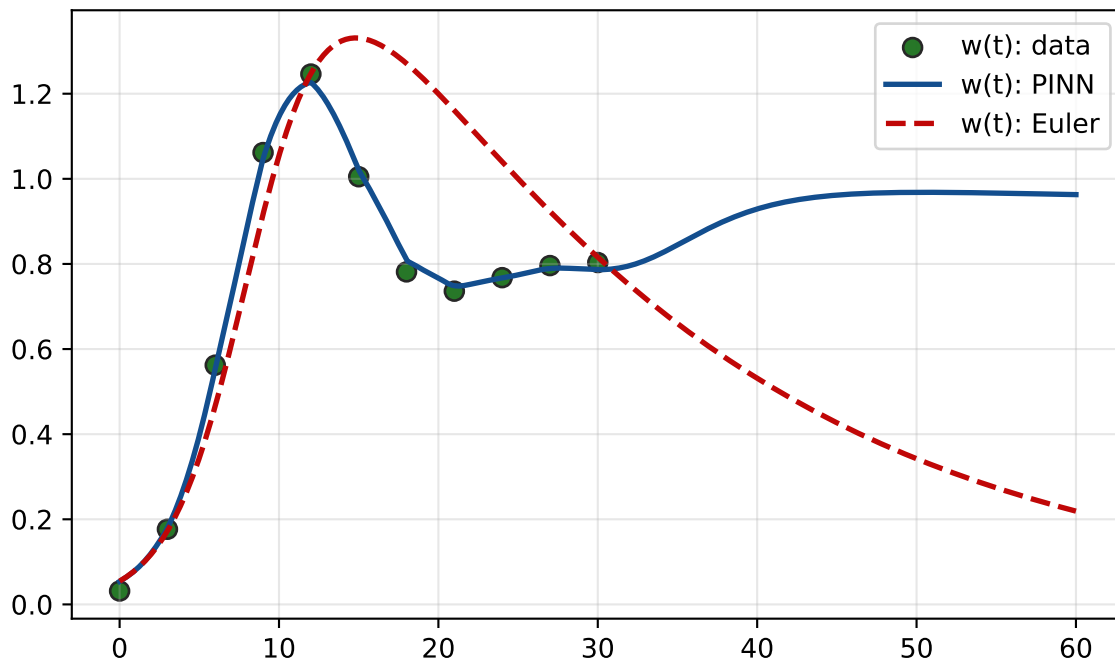
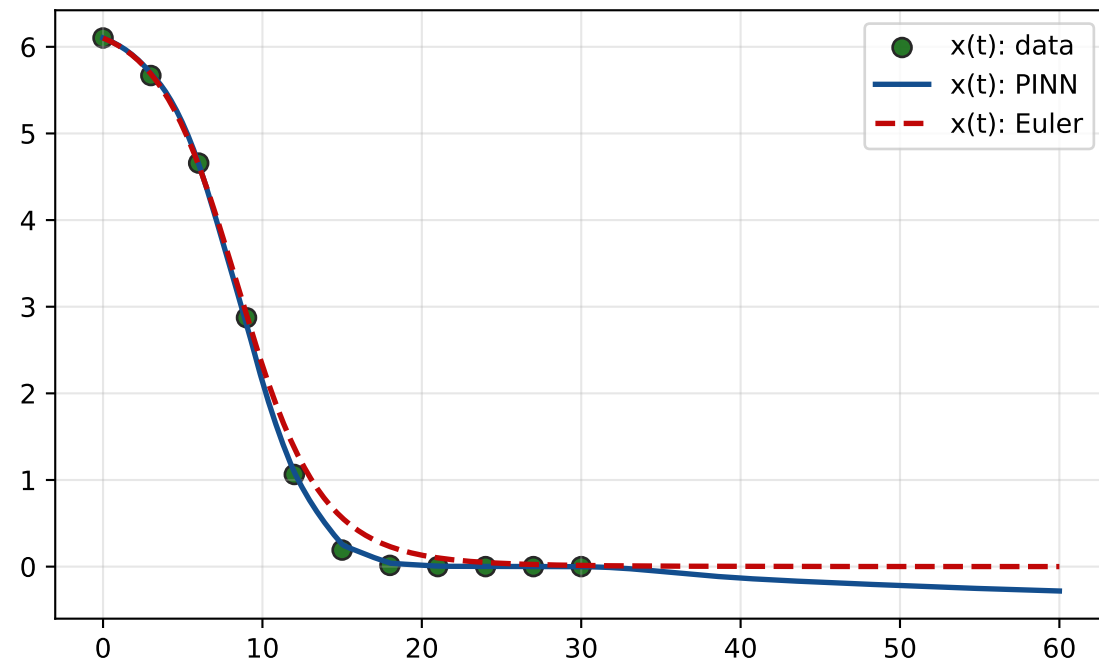


# E7 | PINN vs Euler

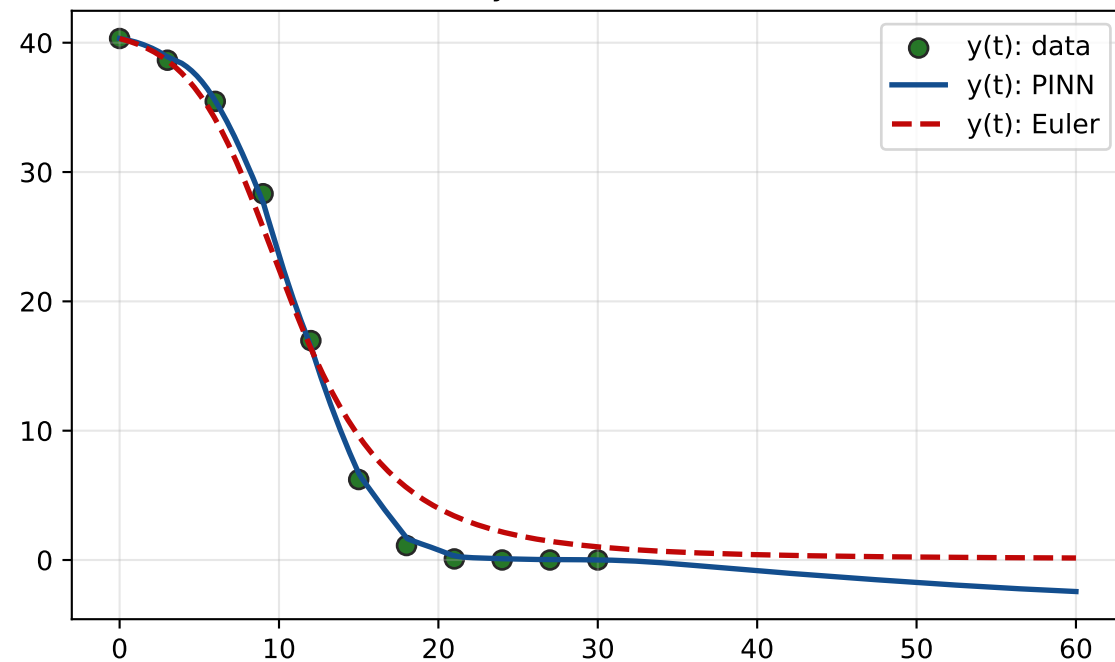
w(t): Biomass



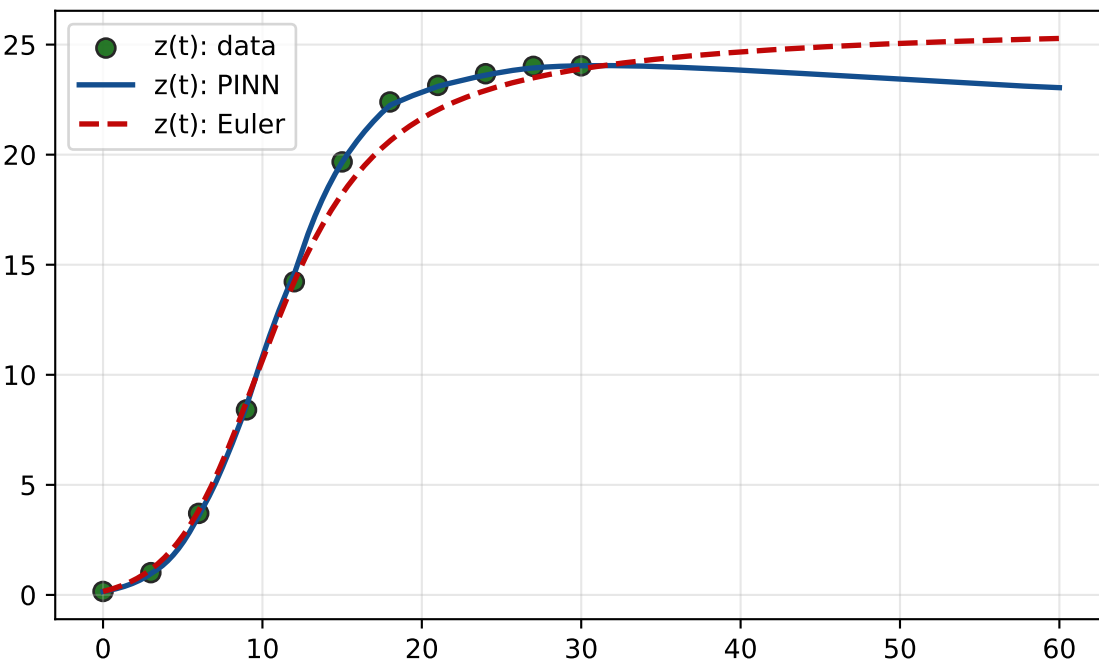
x(t): Glucose



y(t): Fructose



z(t): Ethanol



E7

R2=0.9889, R2\_adj=0.9867, RSS=6.783e+01, AIC=3.615e+01

cond(JTJ)=4.47e+05, used\_pinv=False

| param | estimate     | SE           | MOE_95       | CI_low        | CI_high      | p_value      |
|-------|--------------|--------------|--------------|---------------|--------------|--------------|
| p1    | 8.598588e-03 | 1.690964e-03 | 3.426218e-03 | 5.172370e-03  | 1.202481e-02 | 1.084376e-05 |
| p2    | 9.996920e-02 | 1.092804e-01 | 2.214231e-01 | -1.214539e-01 | 3.213923e-01 | 3.662226e-01 |
| p3    | 1.703294e-03 | 2.461210e-03 | 4.986886e-03 | -3.283592e-03 | 6.690180e-03 | 4.932229e-01 |
| p4    | 2.267719e-01 | 1.151044e-01 | 2.332237e-01 | -6.451781e-03 | 4.599956e-01 | 5.634105e-02 |
| p6    | 1.361717e-01 | 4.799000e-02 | 9.723697e-02 | 3.893470e-02  | 2.334086e-01 | 7.336646e-03 |
| p8    | 6.964741e-02 | 2.519210e-02 | 5.104404e-02 | 1.860337e-02  | 1.206915e-01 | 8.832849e-03 |
| p9    | 2.131691e-05 | 5.923066e-03 | 1.200127e-02 | -1.197995e-02 | 1.202259e-02 | 9.971478e-01 |

| IC             | estimate     |
|----------------|--------------|
| w0 (estimated) | 5.456030e-02 |
| x0 (fixed)     | 6.101894e+00 |
| y0 (fixed)     | 4.032528e+01 |
| z0 (fixed)     | 1.568724e-01 |
| W0 (fixed)     | 0.000000e+00 |